

The questions for the subject "INTERNET TECHNOLOGIES" (Ref. No.: Ex/CSE/PC/B/T/322/2023) are as follows:

Group A (Total Marks: 20) [CO1 and CO2]

Answer ANY TWO questions.

1.
 - a) Decompress and show the complete unabbreviated IPv6 address: AAAA:A:AA::1234.
 - b) What are IPv6 multicast addresses? Describe uses of permanent and transient multicast group addresses.
 - c) What is IPv6 autoconfiguration?
 - d) How does IPv6 handle fragmentation?
2.
 - a) At any time instant local time at host H1 is 1500 and at host H2 is 1200. H1 wants to synchronize its clock with H2 using ICMP. H1 sends an ICMP Timestamp Request message at its local time 1500 and receives the corresponding reply at local time 1600. H2 receives the request at its local time 1300 and sends the reply immediately at local time 1300. Describe the actual messages exchanged (with probable values of different fields, except the checksum) and the steps how H1 adjusts its clock.
 - b) Explain the uses of ICMP router advertisement and ICMP router solicitation messages.
3.
 - a) Assume that, in a TCP connection the initial sequence numbers chosen by the client and the server are 8000 and 9000 respectively. Clearly show values of sequence no., acknowledgement no. fields and flag bits (if any) of segments exchanged (i) During connection establishment (ii) When client sends a segment containing the message "Hello server" and the server answers with a segment containing the message "Hi client" (iii) During connection termination.
 - b) In a connection, the value of cwnd is 3000 and the value of rwnd is 5000. The host has sent 2,000 bytes, which have not been acknowledged. How many more bytes can be sent?

Group B (Total Marks: 10) [CO3]

Answer ANY ONE question.

1.
 - a) Briefly state the objectives of the SSL Handshake protocol.
 - b) Explain how SSL Record protocol secures application data.
 - c) Give examples of two cases when SSL alert protocol messages are generated.
2.
 - a) State the motivation for using design patterns for developing web applications.
 - b) How does a web container provide different outputs for different clients when the same URL is requested?
 - c) State the essential components of a web application framework.

Group C (Total Marks: 30) [CO4]

Answer ANY TWO questions.

1.
 - a) Implement a message passing application using Node.js where a user can send a message to other users who are connected and other users get notified when a user gets disconnected.
 - b) Comment on how the various function calls are invoked at the server side-direct/CPS/callbacks.
 - c) Did you use Event Emitter design pattern? Discuss how it is used.
 - d) State a feature of Node.js/Express that is not supported by Servlets/Spring, Justify.
2.
 - a) Write a Node.JS application that takes two inputs-student name and Course. It collects these inputs from a user and stores them in Mongo DB.
 - b) What is callback hell problem? Discuss 2 strategies to solve it.
 - c) State the pros and cons of Websocket protocol.

3. a) There are two currency classes-INR and GBP. For an online retail store that handles international customers, how can you calculate the total price of a cart with respective indirect taxes (based on respective countries regulations)? Write the basic components of dependency injection for this application and the structure of the Spring Controller class.
- b) How is front controller design pattern utilized here by the Spring framework?
- c) Discuss automatic data marshalling w.r.t Spring framework. Give suitable code snippets.

Group D (Total Marks: 40) [CO5]

Answer ANY FOUR questions.

1. a) What is authorization?
- b) How can you enable authorization check for a URL in Spring? Only write the filter chain part.
- c) How is user authentication invoked in the request-response workflow by the Spring framework?
2. a) A web server hosts a forum. It does not check the comments posted by the users for any security threat. How could this web server pose security threat to another web application hosted at a different server?
- b) Discuss about the session-fixation vulnerability. State how could it be handled using Spring framework.
3. a) Assume that you can only use a hash function H and a symmetric-key encryption algorithm that takes a secret key K. Show how would you transmit a message M from user A to user B so that the following are guaranteed: (i) Message Authentication and Integrity (ii) Confidentiality (iii) Message Authentication, Integrity and Confidentiality.
- b) Discuss the advantages of (i) asynchronous code execution and (ii) synchronous code execution in Node.js with suitable use cases.
- c) How can injection attacks be avoided in Node.js?
4. a) Differentiate between symmetric and asymmetric key cryptography.
- b) Explain how authentication and confidentiality can be achieved using asymmetric key cryptography.
- c) Explain the RSA key generation process.
5. a) What is a secure hash function? What are the properties that a secure hash function must satisfy?
- b) What are the usages of digital certificates? Explain the role of a certification authority (CA).
6. a) Assume users A and B in two different networks. User A owns a public-key certificate (PUB-CERT-A) that is digitally signed by a certificate authority (CA1). CA1's public-key certificate (PUB-CERT-CA1) is digitally signed by another certificate authority (CA2). User A trusts CA1 and User B trusts CA2.
 - i. Explain how you would send a message M from user A to user B such that it provides integrity and authentication.
 - ii. What would be the sequence of steps executed by receiver B to extract the message M?
- b) Assume that you can only use a hash function H and a secret S. Show how would you transmit a message M from user A to user B so that both message authentication and integrity are guaranteed. You should not use any encryption.

Sources:

- [3.2Sem 2023.pdf](#)